

PRODUCT DESIGN BRIEF — SLAB

Portable 6-Channel Production Mixer & USB-C Audio Interface

Revision	C — AS-BUILT, reconciled to the measured Fusion 360 model
Date	11 September 2026
Owner	S. DeFibaugh
Supersedes	Rev A (3 Sep 2026, basis of design) and Rev B
Model of record	SLAB_TOP — 289 bodies, 12 components incl. root, 78 timeline features, 0 named user parameters
Build environment	Autodesk Fusion 360 build 2705.1.11, driven through the fusion360 MCP
Status	CAD complete. Renders, A3 general arrangement and verification report delivered.

What changed in Rev C. Rev A was a basis of design written before CAD. Section 8 of Rev A described a product on a different axis convention, with a different envelope, a different knob count and an 18 mm perimeter bezel. None of that survived contact with the model. This revision restates Section 8 as the measured as-built geometry and records every superseded value in Section 9 so the decision history stays auditable.

0. How to use this document

This brief is written to be executed by an autonomous CAD agent driving Autodesk Fusion 360 through an MCP server, with human review at defined gates.

- **Section 8 is the authoritative geometric definition.** Every figure in it has been measured on the built model, not specified in advance. Where this document and any render, mockup or earlier revision disagree, Section 8 governs.
- **Section 9 is the supersession register.** It records what Rev A asked for and what was actually built, with the reason. An agent must not build anything from the left-hand column.
- **Sections 10–11** define the render, drawing and verification deliverables and the gates that decide whether the model is releasable.
- **Section 12** is the agent execution appendix: the harness, model and MCP configuration needed to run this brief end to end, and the failure modes to expect.
- Any value marked **[TBD-n]** is an open decision. The agent must not invent a value — it builds the stated provisional geometry and surfaces the item at the gate.
- Decisions marked **DEC-C1...C7** are closed. Do not re-raise them.

Reference input: the two concept mockups supplied 3–7 September 2026. They are design intent for form, proportion and CMF. They are AI-generated, are not dimensionally accurate, carry nonsense legend text, and **disagree with each other** on shadow density by 15.6 luminance levels. They must not be traced or photogrammetrically reconstructed. This document governs.

1–7. Product, audio, electronics and power

Carried forward from Rev A without change; none of it is CAD-actionable and none of it is contradicted by the build. Summarised here so this document stands alone.

Section	Substance	Rev C status
1 Product definition	Battery-powered 6-channel mixer / USB-C interface for mobile production. Positioned between a hobby mixer and a rack interface.	Unchanged
2 Design principles	Exposed control deck, one knob per function, no menu diving, field-serviceable battery.	Unchanged — the exposed deck is why DEC-C3 rejects a bezel
3 Features	Channel architecture, EQ, master section, illuminated pads.	Amended: 20 pots (4x5) not 16, per DEC-C2. 10 pads, 3 large knobs.
4 I/O specification	Rear combo jacks, TRS, USB-C; front headphone.	Amended: rear I/O moves to a bolt-in plate per DEC-C4. Left-face USB-C charge ports are built.
5 Audio performance	EIN, THD+N, dynamic range, crosstalk targets.	Unchanged, not CAD-actionable
6 Electronics architecture	Preamp topology, pot selection, PCB stack.	Unchanged. PCB not modelled.

Section	Substance	Rev C status
7 Power and battery	Pack sizing, phantom power budget, user-replaceable pack (legal requirement).	Unchanged. Battery pack and hatch not modelled — carried as forward scope.

8. Mechanical architecture — as-built CAD specification

Every figure below was measured on the built model. Volumes are stated to three decimals because they are checksums: a class that reproduces its volume to four significant figures is correct, and one that does not is localised to a single feature.

8.1 Environment and units

- Autodesk Fusion 360, parametric design type, Capture Design History ON.
- Document **SLAB_TOP**, saved to a cloud project **before the first feature** — the save names the root component, which takes the filename and cannot be renamed through the API afterwards, and it is a precondition for the native drawing.
- The brief states millimetres. **The Fusion API works in centimetres.** Every dimension in this section is mm; divide by 10 at the API boundary.

The model carries zero named user parameters. Rev A required every dimension to be a named parameter or an expression referencing one. That requirement was not met and is recorded as the largest open gap in the model (TBD-14). It is hard-modelled throughout: editing a parameter moves no geometry because there is no parameter to edit. Dimension from measured geometry, never from a parameter table.

8.2 Origin and datum convention — DEC-C1

- Global origin at the geometric centre of the footprint in plan; **Z = 0 at the ground plane**, which the four feet contact.
- +Y is the long axis (340.0 mm) and points away from the operator, toward the rear.**
- +X is the short axis (195.0 mm) and points to the operator's right.**
- +Z is up, normal to the control panel.

DEC-C1 — closed. Rev A §8.2 put the long axis on X (300 mm wide × 180 mm deep). The model is built the other way round and the model is correct: the brief was rewritten to match it, not the reverse. This is not a resize — it moves the rear panel to a different edge of the product and relocates the entire I/O layout. Do not propose rotating the model.

8.3 As-built envelope

Quantity	Value (mm)	Note
Housing X	195.0	short axis
Housing Y	340.0	long axis, DEC-C1
Housing Z	46.0	top face at Z = 46.0 exactly
Corner radius	10.0	all four verticals in plan
Wall	2.5	shell, both mouldings
Split line	Z = 23.0	base 3.5–23.0, upper 23.0–46.0
Foot height	3.5	feet occupy Z 0–3.5
Panel recess	137 × 213	centred (0, +59.5), 2.4 deep from Z 46 to a floor at Z 43.6
Overall bbox	197.040 × 340.040 × 61.210	X −99.520...+97.520 · Y ±170.020 · Z −0.030...61.180

The bounding box exceeds the envelope deliberately: the power toggle overhangs the left end face at X −99.52, and the knobs and pointer inlays stand proud to Z 61.18.

Datasheet envelope is 340 × 195 × 46 mm, not the 64 mm Rev A predicted, because there is no bezel (DEC-C3).

8.4 Build-stage checksums

Confirm each before starting the next class.

Stage	Measured	Derivation
Housing upper, after extrude	1522.926 cm ³	$(195 \cdot 340 - (4 - \pi) \cdot 10^2) \times 23$
Housing upper, after shell 2.5	219.091 cm³	outer – cavity (190×335, R7.5, depth 20.5)
Housing upper, after panel recess	149.056 cm³	– 137×213×2.4, sharp corners
Four edge notches	−1241.3 mm ³	clipped at the ±97.5 outline
Six strip cuts	−87.0 mm ³	6 × 29 × 1.0 × 0.5
Housing base, after shell 2.5	209.947 cm³	depth 19.5, cavity depth 17.0
Two USB-C ports	−190.0 mm ³	2 × 9.5 × 4.0 × 2.5 — clean through the wall
Panel, before shaft holes	56.970 cm³	135 × 211 × 2.0

Stage	Measured	Derivation
Panel Y range	–46.000 ... +165.000	the tell that the panel is at the specified +59.5 and carries its intended 4.0 mm margin at +Y

Face counts are not checksums. SLAB_Housing_Upper measured 63 faces on this build path where an earlier build measured 78, with both volume checksums exact. Face count is a function of feature order. Gate on volume and body count.

Mass is not meaningful. The whole model measures ~3915.8 g at default densities. It is a coarse smell test only and is never quoted as product mass.

8.5 Component tree — 289 bodies, 12 components

Component	Bodies	Extent (mm)	Z range	Appearance
SLAB_Housing_Upper	1	195 × 340 × 23	23.0 – 46.0	SLAB_Gray_Flat
SLAB_Housing_Base	5 (shell + 4 feet)	195 × 340 × 19.5	0 – 23.0	SLAB_Black_Semi
SLAB_Panel_Control	1 (6 faces)	135 × 211 × 2	43.6 – 45.6	SLAB_Panel_Dark
SLAB_Knob_Small	20	Ø13.5 → Ø10.79 × 15.5	45.6 – 61.1	SLAB_Knob_Black
SLAB_Knob_Large	3	Ø32 × 14	45.6 – 59.6	SLAB_Knob_Orange
SLAB_Keypad	10	18.6 sq, R3	44.2 – 46.6	SLAB_LED_Emissive
SLAB_Panel_Ticks	173	graduation marks	45.6 – 46.05	SLAB_Knob_Black
SLAB_Knob_Pointers	23	inlays 1.1×5.6×0.92	58.73 – 61.15	White (20) / Black (3)
SLAB_Power_Switch	2	toggle + LED	44.2 – 59.0	Toggle_Gray / LED_Emissive
SLAB_LED_Indicators	1 (merged)	section dividers	45.6 – 46.1	SLAB_Knob_Orange
SLAB_Panel_Graphics	50	raised graphics	46.0 – 46.15	SLAB_Print_Ink

SLAB_LED_Indicators holds one body, not three — the divider rules touch and merge. SLAB_PCB_Main, SLAB_Battery_Pack, SLAB_Base_Hatch, SLAB_Lightguide and the rear I/O plate are **not built**; they are forward scope (§11).

8.6 Build order — twelve feature classes

One feature class per API call, saving after each. Body counts are exact; the class fails if it lands off-budget.

```
housing upper 1 . base+feet 5 . panel 1 . small knobs 20 . large knobs 3  
pads 10 . ticks 173 . pointers 23 . toggle+LED 2 . dividers 1 . graphics 50 = 289
```

8.7 Housing features

Feature	Qty	Geometry	Position
Panel recess	1	137 × 213, 2.4 deep from Z 46, sharp corners	centred (0, +59.5)
Edge notches	4	1.2 deep into the 2.5 top wall; plan width 21 on X , overhanging the ±97.5 outline so the silhouette steps inward. Heights differ by design.	TL 14.3 @ (−90.2, 143.8) · TR 9.5 @ (91.0, 139.5) · ML 22.9 @ (−90.2, 40.3) · MR 12.4 @ (91.0, 66.5)
Strip cuts	6	29 × 1.0, 0.5 deep ; span panel edge 68.5 to housing edge 97.5	X ±83; Y 112.8 / 57.4 / 12.6
Feet	4	18 × 18, Z 0 → 3.5	X ±77.5, Y ±150
USB-C charge ports	2	9.5 (Y) × 4.0 (Z), cut clean through the 2.5 wall	−X face, Y −38.0 / −16.0, centred Z 13.5

Build the charge ports. PRD Rev B §8 lists them as “recorded as intent, absent from the measured model”. That is stale. They are cuts only, so they change no body count and no checksum notices either way — which is exactly why the two documents drifted. They are also the worked example of DEC-C4 done right: the aperture sits wholly inside one moulding and crosses no parting line.

8.8 Control layout geometry

Small knobs — 20, tapered. Base Ø13.5 (the widest point — evaluate every clearance here), top Ø10.79, height 15.5, Z 45.6 → 61.1, draft 5° **narrowing upward** (a negative taper on a +Z extrude), top edge round **0.75**.

Grid	4 columns × 5 rows = 20 — DEC-C2
X centres	−50.4, −16.8, +16.8, +50.4 (pitch 33.6)
Y centres	+28, +58, +88, +118, +148 (pitch 30.0)
Verification	4 faces per body. Circular edge radii by Z must read [[45.6, 6.75], [60.42, 5.454], [61.1, 4.707]] — base, fillet-meets-cone, fillet-meets-top.

Verify the taper by edge radii, never by mass. A frustum's volume is symmetric in R and r, so an upside-down knob has identical mass, identical body count and an identical bounding box. Radius must *decrease* as Z increases. Three faces instead of four means the fillet did not take, and unfilleted knobs render as flat black holes.

Large knobs — 3, straight. Ø32 × 14, Z 45.6 → 59.6, top edge round 1.10, 4 faces, radii **[[45.6, 16.0], [58.5, 16.0], [59.6, 14.9]]**. Centres X −45 / 0 / +45 at Y −23 (MASTER / PHONES / GLUE).

Pads — 10 lit buttons. 18.6 square, R3 on the four vertical edges, 2.4 tall, Z 44.2 → 46.6 so they stand 0.6 proud of the Z 46.0 face. 10 faces per body.

```
bottom : (-88, -56) (-44, -56) (0, -56) (44, -56) (88, -56)  
left   : (-88, 76) (-88, -4)  
right  : ( 88, 93) ( 88, 40) ( 88, -4)
```

Edge pads sit **0.200 mm** inside the housing outline so they read as flush. That inset is deliberate: a layout check using an arbitrary 1 mm margin produces seven false positives against the correct model. There is no pad recess.

Power switch. Toggle Ø11.0 at **(−94.0, +130.0)**, Z 44.2 → 59.0. Power LED Ø5.2 at (91.0, 139.5), Z 46.0 → 46.6. The toggle bounding box must read **[−99.5, 124.5, 44.2, −88.5, 135.5, 59.0]**.

DEC-C5 — closed. Rev A gave the toggle's X and Z and never stated its Y, while discussing it at length — which makes it read as fully specified when it is not. A rebuild placed it at Y = 0. The value was recovered from the top-down mockup by fitting the ten lit pads' image centroids against their known model coordinates (5.231 px/mm, agreeing to three significant figures on both axes independently) and measuring the toggle in the same frame: Y = +130.2, adopted as **+130.0**. Clearance to the top-left notch is 1.15 mm — know that before nudging either.

8.9 Graduation ticks — 173, all black

Raised **0.45** from Z 45.6, bar width **0.8**, radial, centred on **+Y**, spread over **250°** (angles from -125° to $+125^\circ$).

Ring	Per knob	Radii	Angular step	Count
Small knobs	7	8.75 → 11.25	250/6°	20 × 7 = 140
Large knobs	11	18 → 21.5	25°	3 × 11 = 33

All 173 are black. **SLAB_Tick_Print** (92, 46, 22) is defined and carries zero bodies; the row is kept in the CMF table precisely because it is unused, so the next reader does not “restore” brown ticks as a fix.

8.10 Pointer inlays — 23, two-tone, proud

	Radial offset	Z span	Host top	Proud	Appearance
Small (20)	+3.1	60.23 – 61.15	61.10	0.05	SLAB_Pointer_White
Large (3)	+7.35 (→ Y –15.65)	58.73 – 59.65	59.60	0.05	SLAB_Knob_Black

Boxes 1.1 (X) × 5.6 (Y) × 0.92 (Z), pointing +Y. **Identify them by position, not name:** the large-knob inlays are the only pointers below Y = 0.

No applied marking may be coplanar with its host face. Built at exactly Z 59.60 against a 59.60 knob top, the large-knob pointer rendered on one knob out of three with identical and correct material on all three. Clear every inlay by ≥ 0.05 mm.

8.11 Section dividers — one merged body

Orange, raised **0.5** from Z 45.6. One horizontal rule 132×0.9 at Y +4, and two verticals 0.9×47 at X ± 22.5 running **downward** from Y +4 to Y –43, into the gaps between the large knobs. They bracket the master row as a group; they do not box each knob individually. All three touch, so five sketch profiles extrude into **one** body with bounding box [–66.0, –43.0, 45.6, 66.0, 4.45, 46.1].

8.12 Panel graphics — 50 bodies, five sub-sketches

All raised **0.15** from Z 46.0, on the lower deck (Y < –46), outside the panel.

Sub-class	Bodies	Geometry	Profile rule	Signature
Lower-face table	1	3 rules 193×1.5 at Y –67.1 / –93.6 / –116.2; divider 1.0×49.1 at (–23.8, –91.65); seam 1.0×113 at (+68.5, –103.5)	keep profiles with min bbox dim ≤ 2 mm	top face 1023 mm² (21 profiles offered: 19 bars + 2 cells)
Grid blocks	2	X –8.3→5.2 and 16.8→30.3, Y –141→–127.6; 4 h-bars + 4 v-bars at width 0.8, inset so outer edges land on the outline (h centres –140.6...–128.0 step 4.2; v centres $x0+0.4...x0+13.1$)	same min-dimension rule	75.8 mm² each
Scale ticks	30	X –56.6 → –14.1 (spacing 42.5/29), baseline Y –146.7, width 0.6, length 3.6 on every fifth else 2.4	keep all	2.16 mm ² each
Legends R, I, V	3	“R” at (–69.5, –146.7), “IV” at (60.2, –146.7); 9.5 high, stroke 1.6	centroid-in-region , dropping only the R’s counter	R 42.1 · I 15.2 · V 27.9 mm ²
Waveform	14	14 bars width 0.8, X 34.0 → 62.0 (spacing 28/13), centred on Y –134.3, heights varying inside the –141...–127.6 band	keep all	≤ 10.7 mm ² each

The min-dimension profile filter belongs to orthogonal bar grids only. It correctly drops the enclosed cells in the table and the grids, where every real element is a 0.8–1.5 mm bar. Applied to the legends it also deletes the R’s leg and both of the V’s strokes — their bounding boxes are 2.5–3.3 mm at the narrow end — and the model builds a “P” and a collapsed V *while still reporting the correct body count of 3*. The discriminator is face area, not body count.

TBD-17 — the waveform is not itemised in the PRD. It is visible in both mockups and is required to reach the specified 50 graphics; its position and 14-bar form are this build’s interpretation. Itemise it or replace it.

8.13 CMF — ten appearances, 289 bodies

Configure all ten **while unapplied**, then assign. Writing a property on an applied appearance forks a local copy; one session reached 167 appearances, 156 of them duplicates of a single emissive material. Set **surface_albedo** and **opaque_albedo** **together** — the renderer reads surface_albedo, and writing only opaque_albedo leaves a stale colour that no geometric checksum sees.

Appearance	Albedo	Rough	f0	Bodies	Applied to
SLAB_Gray_Flat	196,196,196	0.50	0.040	1	upper housing
SLAB_Black_Semi	48,48,50	0.38	0.055	5	base + 4 feet
SLAB_Panel_Dark	78,78,82	0.60	0.042	1	control panel
SLAB_Knob_Black	42,42,44	0.30	0.058	196	20 small knobs + 3 large pointers + 173 ticks
SLAB_Knob_Orange	252,108,64	0.34	0.050	4	3 large knobs + dividers
SLAB_LED_Emissive	255,122,48	0.26	0.050	11	10 pads + power LED
SLAB_Print_Ink	58,58,62	0.70	0.035	50	panel graphics
SLAB_Pointer_White	212,212,212	0.50	0.045	20	small-knob pointers
SLAB_Toggle_Gray	200,200,200	0.35	0.050	1	toggle
SLAB_Tick_Print	92,46,22	0.62	0.040	0	defined, deliberately unused

SLAB_LED_Emissive additionally carries opaque_emission = True, **opaque_luminance = 150.0**, opaque_luminance_modifier = (255, 140, 70). Confirmed independently: it renders the pads at rgb (251.7, 141.8, 64.7), G:R 0.564. The pads are light sources and must not be approximated with glossy orange.

End state: **11 appearances** (the ten SLAB_* plus the library source they were copied from), zero duplicates, zero stray face overrides. Resolve appearances by iterating the collection — itemByName returns only the first of a duplicated name.

8.14 Layout checks — expected values

Evaluate knob clearances at the widest section (Ø13.5 / Ø32). A correct model returns zero violations and these exact figures.

#	Check	Correct model	Limit
1	Pairwise knob gap	13.000 mm (Knob_Lg_1 ↔ Knob_Lg_2)	≥ 4.0
2	Knob vs pad clearance	7.700 mm (Knob_Lg_1 ↔ Pad_05)	no overlap
3	Panel inside the recess	1.0 mm all four sides	≥ 0
4	Pad inset from housing outline	0.200 mm	≥ 0 (0.2 deliberate)
4b	Pad-to-panel gap	0.700 mm	> 0
5	Indicator not buried under a knob	—	NOT IMPLEMENTED
6	Boss-to-aperture ≥ 6.0	—	NOT IMPLEMENTED — no bosses built

Checks 5 and 6 are not silently passing. Report them as unimplemented. Re-run the whole set after any *dimension* change, not just a position change: Ø30 → Ø32 broke a 4 mm clearance, and Ø20 → Ø13.5 changes which pair governs the worst case.

8.15 The governing constraint — DEC-C3

Rev A §8.4: no control may protrude above the enclosure plus 2 mm.

Enclosure_Top_Z	46.000 mm	(housing upper top face)
Tallest_Control_Z	61.180 mm	(small-knob pointer inlay, bbox Z max)
46.000 - 61.180	= -15.180 mm	
required	>= +2.000 mm	
SHORTFALL	17.180 mm	

DEC-C3 — closed. Not blocking. Rev A's own proposed fix was an 18 mm perimeter bezel raising the rim to 64 mm. That would have destroyed the exposed control deck that is the product's identity (§2). **Bezel_Height stays 0** and the backpack requirement is met by a transport cover accessory, not by enclosure geometry. Report the arithmetic at every gate; do not flag it as blocking and do not add a bezel. PM and ID signature remain outstanding before tooling release (TBD-12).

9. Rev A → Rev C supersession register

The left-hand column is **not to be built**. It is recorded so the decision history stays auditable.

Rev A said	Rev C as-built	Why
Overall_Width 300.0 (X), Overall_Depth 180.0 (Y) — long axis on X	X 195.0, Y 340.0 — long axis on Y	DEC-C1. The model is correct; the brief was rewritten. Moves the rear panel to a different edge.
Bezel_Height 18.0; envelope 300 × 180 × 64	Bezel_Height 0 ; envelope 340 × 195 × 46	DEC-C3. A bezel destroys the exposed deck. Met by a transport cover.
Split_Line_Z 14.0	23.0	Gives a roughly 50/50 two-tone and lets the charge ports sit wholly in the base band.
Corner_Radius 8.0	10.0	As built; softer plan silhouette matching the mockups.
4 × 4 knob matrix + 2 = 18 small knobs	4 × 5 = 20	DEC-C2. 20 pots governs. Do not correct the count downward.
Knob_Sm_Dia 15.0, straight, height 14.5	base Ø13.5 → top Ø10.79 , 5° taper, height 15.5, R0.75 top	Ø20 straight cylinders were built, judged and rejected as squat and crowded. The taper is the form.
Knob_Lg_Dia 30.0, height 15.5	Ø32.0 , height 14.0, R1.10 top	As built. Ø30→Ø32 is what makes the worst knob gap 13.000.
Knob_Pitch_X 34.0	33.6	As built.
Matrix origin (−112.0, +38.0); large knobs on a diagonal at (+100,+34), (+82,+4), (+64,−26)	small-knob grid X ±50.4/±16.8, Y +28...+148; large knobs in a row at X −45/0/+45, Y −23	Consequence of DEC-C1 and the 195 mm short axis. The Rev A X values fall outside the housing.
Pad 18.0 × 12.0, R2.0, proud 0.8; two rows of five at Y −52 / −68, 44.0 pitch	18.6 square, R3.0, proud 0.6 ; five along the bottom at Y −56 plus two left and three right	As built. Perimeter placement, not two lanes.
Panel 268 × 148, Panel_Inset 0.4	135 × 211 × 2.0 in a 137 × 213 recess, floor Z 43.6	Consequence of DEC-C1. 1.0 mm margin all round.
Combo_Cut_Dia 24.0, 4 rear combo apertures + TRS, cut into the moulding wall	No aperture in either moulding wall. Rear I/O on a bolt-in plate in a recessed bay	DEC-C4. A Ø24.0 aperture fits neither band (19.5 / 23.0 mm) — it cannot be placed without crossing the Z 23.0 parting line.
Power toggle: X −94.0 and Z given, Y never stated	Y = +130.0	DEC-C5. Recovered photogrammetrically from the pad centroids in the top-down mockup.
50 named user parameters driving all geometry	0 named user parameters	Not met. Recorded as TBD-14, the largest open gap in the model.
Battery hatch, PCB, lightguides, feet detail, internal ribs, draft analysis	Not modelled	Forward scope — see §11.

10. Visual and 2D deliverables

10.1 Exposure

Scene exposure is a property of the **document**, not the product — the same product with byte-identical CMF needed 22000 lux in one document and 1700 in a from-scratch rebuild. Derive it every run, in one pass: set brightness = **1700**, fire one 256 px quality-25 render, and measure the housing class. **177–185 is converged**. Measured: 184.5.

Measure against a rendered plate, not a colour guess. The backdrop renders perfectly uniform (std 0.000 at rgb 178,178,178), so hiding every occurrence and rendering the empty plate gives an exact difference matte. At a correct exposure the grey housing sits within ~8 levels of the plate, so a colour-distance key composites the product half-transparent.

10.2 Cameras and the two finals

Deliverable	Camera	Render	Composite
Hero	perspective, azimuth 303°, elevation 46°, fov 14°, eye d = 152 cm from target (0, 0, 26 mm), up +Z	quality 95, height 1200, view='current'	--aspect 1.328 --bloom-gains 0.10,0.055
Plan	orthographic, eye +Z, up +Y, fit then viewExtents × 1.10	quality 95, height 1400, view='current'	--aspect 0.656 --plan --bloom-gains 0.08,0.04

start_render defaults to view='iso' and applies its own camera preset, discarding the live viewport camera. The calibrated cameras above only reach the raytracer when view='current' is passed. The return payload states which view it used — read it. Also: width is discarded; output is height × live viewport aspect.

Fusion rescales a camera's eye to preserve viewExtents. A requested 152 cm comes back as 247.6 cm — direction preserved exactly, a pure 1.629× scaling. Push the camera, read the achieved distance, scale viewExtents by (target/actual)², push again. Converges in one iteration.

10.3 Acceptance — two parts, not one

Test	Accept	Measured this build
LED pad G:R ratio	0.55 ± 0.03	hero 0.564 · plan 0.566
LED pad luminance	inside 144.9 – 160.5	159.7 / 160.0
Ten-pad connected components	exactly 10 blobs, per-channel spread ≤ 7 levels	10 blobs; [2.7, 6.3, 2.9] hero, [6.3, 7.4, 2.4] plan
Housing luminance	within ~2 levels of reference	185.5 / 185.4 vs references 174.9 and 179.5 — reported as a gap, not chased

The G:R half is not optional. The green channel clips first, so a blown pad reads G:R 0.90 while its luminance still looks plausible — luminance alone will accept it, and an eight-pass loop once converged on exactly that. The ten-pad blob check is the only test that catches a one-of-N defect such as a dead pad from a stray face override; a correct model measures a spread of 2–7 levels, the known defective one measured 47.

The two mockups disagree with each other — by 4.6 levels on housing and 15.6 on the dark class. Converging on the midpoint moves you toward one and away from the other. Land inside the band and stop. Budget two finals per view; if the loop is still moving after that, report the gap with the number attached.

10.4 General arrangement drawing

Line-art views are captured with explicit orthographic cameras at one shared viewExtents, then cropped analytically. **Expect a crop-aspect error ≤ 0.11 %** (measured: top 0.04, front 0.07, right 0.11). Anything over 1 % means a clipped capture — fix that before placing a single dimension.

The sheet is A3 ISO at 1:2.5: three dimensioned orthographic views, an isometric with balloons, a 12-row parts list totalling 289 bodies, and notes. Rasterise the result and look at the whole page — layout collisions are invisible in the PDF object model.

Do the native drawing LAST. create_drawing(sheet_size='A3ISO', standard='ISO', units='mm', center_marks=False, open_drawing=True) works on build 2705.1.11, but it exceeds the MCP request timeout: it returns *Request timed out* and ping stops answering for ~45 s. Do not retry — poll ping, then confirm with get_design_type. Once a Drawing document is active, execute_code no longer runs and **only a human clicking the Design tab can return the session to the model. Never call export_drawing_pdf** — it opens a modal and hangs the MCP server.

11. Decision registers

11.1 Closed — do not re-raise

ID	Decision	Ruling
DEC-C1	Axis convention	Long axis Y (340), short axis X (195), +Y away from the operator. Do not propose rotating the model.
DEC-C2	Knob count	20 pots (4 × 5) governs. The $4 \times 4 = 16$ in Rev A is superseded. Do not correct the count downward.
DEC-C3	The bezel	Not built. Bezel_Height stays 0; the 17.180 mm shortfall is met by a transport cover accessory. Report the arithmetic; do not flag it as blocking.
DEC-C4	Rear I/O	No connector aperture may cross the Z 23.0 parting line. Rear I/O is carried on a bolt-in plate in a recessed bay; the bay may span the split, the apertures are cut in the plate.
DEC-C5	Power toggle Y	Y = +130.0.
DEC-C6	The keypad	The 10 pads are lit buttons — `SLAB_LED_Emissive` with emission on, luminance 150.0, modifier (255,140,70). They are light sources, not printed ink, and must not be approximated with glossy orange. The same appearance carries the power LED: 11 bodies.
DEC-C7	Section-divider direction	The two verticals run DOWNWARD , from Y +4 to Y −43, into the gaps between the large knobs at Y −23. An earlier build ran them upward to Y +51, away from the knobs. Re-apply the appearance after any rebuild of this body — it has returned as `Body2` on a library default before.

DEC-C4 regression tell. In SLAB_Housing_Upper and SLAB_Housing_Base, count cylindrical faces with axis (0, −1, 0) whose extent touches $Y \approx \pm 170$. **The correct model has zero.** Because +Y is the edge away from the operator, a plan view reads such apertures as holes in the top of the deck — look at the rear face. Mass will not catch it: removing the offending cuts moved the whole-model figure by 38.973 g and the body count not at all.

11.2 Open — carried forward

ID	Item	Owner	Due
TBD-12	PM + ID signature on DEC-C3 (no bezel, transport cover) before tooling release	PM / ID	pre-tooling
TBD-13	Rear I/O plate and recessed bay — not modelled. HOLD-A (Ø24.0 combo jack) is a geometry problem, not only a sourcing question	ME	DVT
TBD-14	Zero named user parameters. The model is hard-modelled throughout; no dimensional change can be made parametrically. Largest open gap	CAD	before any dimensional change
TBD-15	Panel shaft holes — 6 faces, a plain slab. Shaft Ø unstated everywhere; volume arithmetic implies $\approx \text{Ø}8.3$ through 2 mm	ME	P2
TBD-16	Boss positions predate the axis correction (X ± 150 falls outside a 195 mm housing). Boss-to-aperture check unimplemented	CAD	P2
TBD-17	Waveform graphic (14 bodies) not itemised in the PRD; position and form are this build's interpretation	ID	pre-drawing
TBD-18	Legend glyph construction (stroke 1.6, R/I/V forms) unspecified — the brief gives position and 9.5 mm height only	ID	P2
TBD-19	Scale-tick anchoring — “at Y −146.7” does not say baseline or centreline. Built as baseline, extending +Y	ID	P2
—	Not modelled: orange panel speckle, in-panel legend text, internal ribs, draft angles and draft analysis, PCB, battery pack, base hatch, lightguides	—	—

12. Agent execution appendix — Hermes + Nemotron

This brief was executed end to end by an agent driving Fusion 360 through the **fusion360** MCP server. This section is what a different harness needs in order to reproduce it.

12.1 The stack, and what is portable

Layer	What it is	Portable?
Fusion 360 MCP server	A Fusion add-in at %APPDATA%\Autodesk\Autodesk Fusion 360\API\AddIns\Fusion360MCP, exposing ~110 tools. Runs <i>inside</i> the Fusion process.	Yes, unchanged. MCP is a client-side protocol; the server has no idea which model is on the other end.
Skills	Six SKILL.md procedure files (slab-run, slab-product-spec, fusion-scratch-build, fusion-mcp-driving, fusion-photoreal-render, fusion-2d-drawings), ~23.8k tokens total	Yes. Hermes Agent uses the open agentskills.io format — the same frontmatter + markdown shape.
Helper scripts	composite.py, make_views.py, make_drawing.py, slab_sheet.json, measure_classes.py	Yes , but pin them to a stable path — see R7.
Model	Any model capable of long-horizon tool use and long code-in-JSON arguments	Nemotron 3 Ultra ranks at or near the top of open-weight models on BFCL and τ -bench.

The MCP is not the problem. Hermes reads the same config shape as Claude Code — the mcpServers block in ~/.claude.json maps to mcp_servers in ~/.hermes/config.yaml, and hermes import-agent claude-code migrates the MCP servers, skills and instructions in one command. That is the fastest path.

12.2 Minimum configuration

```
# ~/.hermes/config.yaml
mcp_servers:
  fusion360:
    command: "<same command/args as your Claude Code mcpServers entry>"
    tools:
      include:                # 8 tools, not 110 -- see R3
        - ping
        - get_design_type
        - execute_code
        - render_view
        - start_render
        - get_render_status
        - set_scene_environment
        - create_drawing
```

Model selection: /model nvidia/nemotron-3-ultra:free in a session, or any OpenAI-compatible NIM / vLLM endpoint. If you drive Hermes through the NVIDIA NeMo Fabric adapter (nvidia.fabric.hermes), three of its defaults will break this workflow — see R2 and R5.

12.3 Failure modes to expect, ranked

#	Failure	Symptom	Fix
R1	Hermes runs somewhere Fusion isn't	MCP server unreachable. Hermes is designed to run on a VPS, Daytona, Modal or SSH backend; this workflow must reach a live Fusion GUI.	Run Hermes on the same Windows machine as Fusion , with the local terminal backend. A remote backend cannot drive this.
R2	max_tokens too small for a build call	Truncated Python, syntax errors mid-class, or the model silently emitting a smaller class.	The NeMo Fabric Hermes adapter defaults max_tokens: 512. A single feature class is 150+ lines of Python. Set ≥ 8192 .
R3	Tool-list flooding	Wrong tool chosen; hallucinated tool names. ~110 Fusion tools on top of Hermes' 60+ built-ins is ~170.	Use the tools.include list in 12.2. This workflow needs 8 .
R4	No vision	The one-second viewport capture that catches a "P" built instead of an "R" cannot be read.	Substitute the numeric gates in 12.4. Confirm whether your Nemotron variant is multimodal before relying on any visual gate.
R5	Agent loop budget too low	Run stops mid-build with no error.	~50 dependent tool calls over ~45 min. Set runtime.max_turns / AI-Agent.max_iterations to ≥ 150 . Set reasoning effort to medium or high — the Fabric adapter defaults to none, and the closed-form checksums need it.
R6	MCP call timeout	<i>Request timed out</i> on create_drawing and on the 289-body appearance pass — both of which had succeeded .	Raise the MCP tool timeout to ≥ 180 s . And never blind-retry a non-idempotent call: re-query state first.

#	Failure	Symptom	Fix
R7	Unstable paths	Scripts not found; renders not found.	Fusion writes renders into a build-hash directory that changes on every Fusion update. Read the path from the start_render return payload — do not hardcode it. Copy the five helper scripts to a stable folder and reference that.
R8	Command-approval prompts	Run stalls waiting for a human.	An unattended 45-minute run needs approval policy configured up front, or it will block on the first terminal call.
R9	Fusion parked in a Drawing document	Every execute_code returns <i>'Drawing' object has no attribute 'designType'</i> .	A human must click the Design tab. Begin every run by calling get_design_type and stopping if it errors.
R10	SLAB_TOP already exists	A second run's saveAs versions or collides with the existing cloud document.	Decide up front: new name, or accept versioning. The gate in 12.4 will otherwise pass against the <i>old</i> model.
R11	Missing Python packages	Composite and drawing steps fail.	The helper scripts need PIL, numpy, reportlab , and pdftoppm (poppler) for the raster check. They do not need scipy — labelling is done in numpy.

12.4 Numeric substitutes for the visual gates

If the model cannot see images, every gate below still works. These were sufficient to catch every defect found in this build except one, and that one has a numeric substitute too.

Visual gate	Numeric substitute	Pass value
"Are all the bodies there?"	body count per component	289 total, 12 components
"Is the knob the right way up?"	circular edge radii by Z	[[45.6, 6.75], [60.42, 5.454], [61.1, 4.707]], decreasing
"Is the table filled in?"	Gfx_Table top face area	1023 ± 10 mm ²
"Does the legend read R, or P?"	legend body areas — a legless R still reports 3 bodies, so body count cannot see it	R 42.1 · I 15.2 · V 27.9 mm² (±2)
"Is the panel the right colour?"	resolved appearance audit through the bodies	0 mismatches, 0 bodies without appearance, 0 stray face overrides
"Are the pads lit evenly?"	ten-pad connected components on the render	10 blobs, per-channel spread ≤ 7
"Are the pads blown out?"	G:R ratio	0.55 ± 0.03
"Is the exposure right?"	housing class luminance on a 256 px pass	177 – 185
"Are there holes in the deck?"	cylindrical faces with axis (0,−1,0) touching Y ±170	0
"Is the layout sane?"	the four implemented layout checks	13.000 / 7.700 / 1.0 / 0.200 — zero violations
"Is the drawing clean?"	crop-aspect error per view from make_views.py	≤ 0.11 %

12.5 A note on the MCP add-in itself

Two behaviours in the add-in are worth fixing upstream rather than working around:

- **The wrapper reads design.designType unconditionally** for its before/after instrumentation. A Drawing product has no such attribute, so every tool dies at the wrapper before its handler runs — which is why a Drawing document is a one-way door. Guarding that read would make the drawing stage recoverable.
- **The request timeout is shorter than some legitimate operations.** create_drawing and a 289-body appearance pass both exceeded it while completing normally. Because a timeout is also the signature of a modal-dialog hang, the two are indistinguishable to the agent — the only safe response is to poll ping and re-query state, never to retry.

Neither is a Nemotron or Hermes problem. Both are present today with any client.

Document	SLAB_PRD_RevC — as-built product design brief
Model of record	SLAB_TOP · 289 bodies · 12 components · 78 features · 0 parameters

Companion files	SLAB_RevC_agent_prompt.md (paste-ready run prompt) · SLAB_GA_RevC.pdf · slab_hero.png · slab_top.png · SLAB_RevC_verification_report.md
Generated	11 September 2026
